Optical response of a polished plastic scintillator read out by side-mounted SiPMs: a Geant4 matrix study of reflector and layout effects

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Abstract

We present a Geant4 (v11.3.2) optical-transport study of a 250 mm square, 10 mm thick polished plastic scintillator read out by side-mounted silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. Four SiPM layouts—8, 4 and 2 directly coupled sensors, and a single SiPM reading a wavelength-shifting (WLS) fibre clamp—are combined with four non-instrumented-face treatments: a bare plate in a black holder, TiO_2 paint, aluminium foil with grease, and high-gloss paper. A matrix of three 50 mm square beam patches is complemented by a uniform whole-plate irradiation, giving 5.2×10^5 events in total. The simulation uses the Geant4 unified surface model, digitized scintillator and Kuraray Y-11 optical tables, a digitized ASD-NUV4S-P-40 photon-detection-efficiency curve, and boundary-process photodetection. Glossy paper is the brightest and most uniform wrapper in every layout; TiO_2 is the most position-dependent and the weakest wrapper in the three direct-coupled layouts. The bare plate is *not* the weakest collector: with a true 1.2 mm air gap it is a near-lossless total-internal-reflection guide and is second only to glossy paper in the direct-coupled layouts, whereas the reference measurement gives it the lowest light yield. The WLS layout collects the fewest photoelectrons and reverses the aluminium/TiO_2 order. We quantify the position contrast over the full plate and the arrival-time and wavelength structure of the detected light, and discuss the surface or contact loss that would be required to reproduce the measured bare-plate suppression. The results provide design guidance and identify the plate–holder interface as the main unresolved ingredient.

1. Introduction

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers, veto systems and cosmic-ray muon tomography [1, 2]. The light yield of such a detector is governed less by the scintillator itself than by the treatment of the non-instrumented faces. A polished plate surrounded by air traps light by total internal reflection (TIR) and can act as a nearly lossless light guide, whereas a plate wrapped in an absorbing or diffuse material loses photons at every bounce. The competition between these mechanisms is a multi-bounce photon random walk that is conveniently studied with Geant4 optical transport [3, 4, 5].

Rao *et al.* [6] recently characterised a 250 mm×250 mm×10 mm UPS-92S plate read out by 4×4 mm^2 SiPMs in four layouts and with four reflector treatments, using a 50 mm×50 mm muon coincidence detector to define centre, edge and corner regions. Their pulse-height spectra show glossy paper and aluminium foil as the brightest wrappers, TiO_2 paint as an intermediate case, and the bare plate in its black holder as by far the weakest. The present work asks how much of that ordering follows from the stated geometry and optical properties, and what the answer implies for the design of similar detectors.

We build a self-contained Geant4 model of the same plate, SiPMs, wrappers and WLS-fibre clamp, and run a matrix of four layouts × four wrappers × three beam patches, plus a uniform whole-plate irradiation. The paper has three goals: (i) to document the optical model and its inputs at a level that allows the results to be reproduced; (ii) to establish which experimental orderings are robust predictions of the stated construction and which are not; and (iii) to translate the results into practical guidance on wrapper choice, layout and read-out.

The main result is a qualified reproduction. Glossy paper is the brightest and most uniform wrapper, and TiO_2 is the most position-dependent and the weakest wrapper in the direct-coupled layouts, in agreement with the measurement. The bare plate, however, is predicted to be the *second*-brightest wrapper, not the weakest, because a polished plate separated from its holder by an air gap is an efficient TIR guide. Reproducing the measured bare-plate suppression requires an additional loss channel at the plate–holder interface, either optical contact (frustrated TIR) or a substantially rougher surface than the nominal polished finish. We discuss the scale of that loss and its experimental signatures.

The paper is organised as follows. Section 2 describes the detector model,

the surface treatments and the optical-physics implementation. Section 3 defines the simulation campaign and the efficiency observables. Section 4 presents the yields, spectra, spatial response, uniformity and timing. Section 5 discusses the bare-plate discrepancy and the limitations of the model, and Sec. 6 concludes.

2. Detector model

2.1. Scintillator, SiPMs and layouts

The active volume is a 250 mm×250 mm×10 mm polyvinyl-toluene (PVT) plate. It uses the Geant4 NIST material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical-property table built from the EJ-200 and UPS-923A data sheets [7, 8, 9]. The refractive index is $n = 1.58$, the light yield is $1.0 \times 10^4\,\mathrm{MeV}^{-1}$, the emission peak is 425 nm, the bulk attenuation length peaks at 3.8 m and falls in the ultraviolet, the scintillation decay and rise times are 2.1 ns and 0.9 ns, and the Birks constant is $0.126\,\mathrm{mm\,MeV}^{-1}$ [10, 11]. The density is $1.032\,\mathrm{g\,cm}^{-3}$ from the Geant4 NIST database. These choices are collected in Table 1; the unusual bulk-attenuation shape is a modelling choice that mimics wavelength-shifter self-absorption in the ultraviolet while keeping the visible range transparent.

Each SiPM [12] is represented mechanically as a $4 \times 4 \times 0.6$ mm box of epoxy ($n = 1.55$), coupled to a side face through a 0.1 mm silicone-grease pad ($n = 1.46$). The photocathode is a `dielectric_metal` boundary whose efficiency is the digitized ASD-NUV4S-P-40 photon-detection efficiency (PDE), peaking at 0.431 near 410 nm [13]. The four layouts are shown in Fig. 1:

Config 1 (8 SiPMs): two SiPMs on each lateral face, at ±50% of the half-length.

Config 2 (4 SiPMs): one SiPM per face, at the same offsets as config 1; the removed slot is rotated from face to face so the layout remains four-fold symmetric.

Config 3 (2 SiPMs): both SiPMs on one narrow face, at ±50%.

Config 4 (1 SiPM + WLS): a single SiPM reading a WLS-fibre clamp along one narrow face.

In the report frame of Ref. [6], used throughout this paper, $+y$ is the top edge and $+x$ the right edge; configs 3 and 4 are on the top edge. Simulation

coordinates are mapped to this frame by a 90° clockwise rotation, $(x, y) \to (y, -x)$, before plotting. The rotation is rigid and leaves all scalar results unchanged.

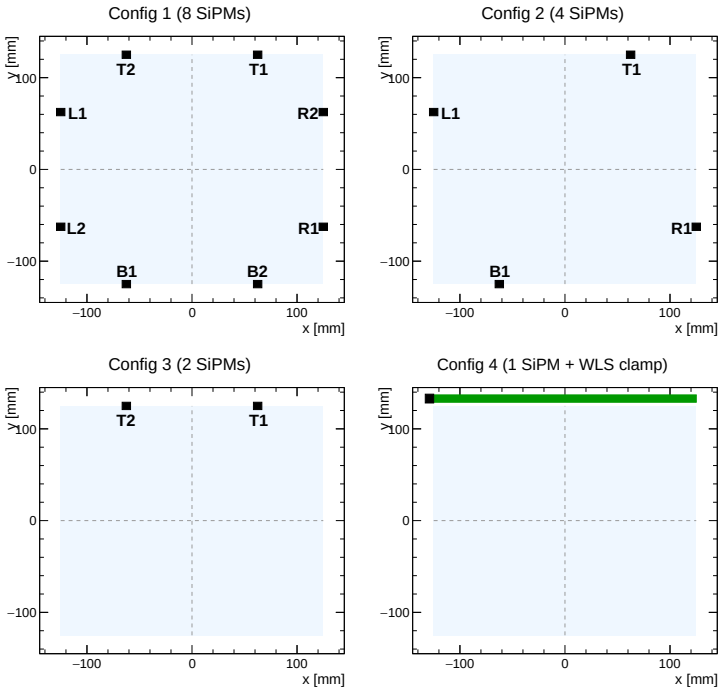


Figure 1: Plan view of the four layouts on the 250 mm×250 mm plate. Black squares mark the SiPMs (oversized for visibility); the dashed lines are the $x = 0$ and $y = 0$ axes. Config 1 uses eight sensors, config 2 four (one per face, at the config-1 offsets), config 3 two on the top edge, and config 4 a single SiPM at the left end of the WLS-fibre clamp on the top edge. The labels T/B/R/L denote the four faces and the numerals the two offsets on each face.

2.2. Reflector treatments

The four treatments are applied as logical skins on the non-instrumented surfaces; the grease/SiPM patches override the skin locally through border surfaces. Table 2 lists the parameters actually attached in the code. The mapping is the heart of the model because it determines the light-collection ordering.

Bare plate in a black holder.. No skin is attached to the plate. The native polished scintillator–air boundary ($n = 1.58$ to 1.0003) is separated from the black holder by a real 1.2 mm air gap. Light below the critical angle $\theta_c = \arcsin(1/1.58) = 39.3°$ refracts across the gap and is absorbed by the holder, whose skin has $R = T = 0$ and whose bulk absorption length is 1 µm. Light above θ_c is returned by TIR. The SiPMs are still coupled through grease pads, so the plate is not isolated from the read-out. This is the least constrained configuration: it assumes the plate is everywhere decoupled from the holder. In configs 1–3 the black holder is built; in config 4 the clamp replaces the holder on the instrumented face, so the "bare" label there means only that no reflector skin is applied.

TiO_2 paint.. The plate is coated directly with a white diffuse paint, modelled as `groundfrontpainted`: a random draw against $R(\lambda)$ returns the photon with a Lambertian distribution and no specular or TIR component. The reflectance rises from 0.955 in the ultraviolet to 0.983 near 470 nm. This is the physically correct description of a thick scattering paint, but the absolute curve is a fitted white-paint representative, not a measurement of EJ-510.

Aluminium foil with grease.. The foil is modelled as `polishedbackpainted` with the backing index set to the coupling grease ($n = 1.46$). The scintillator–grease interface is a Fresnel/TIR boundary with a wide critical angle,

$$\theta_c = \arcsin(1.46/1.58) = 67.5°,$$

so most light refracts into the grease and is returned by the foil as a pure specular reflection. The reflectance is the normal-incidence value of evaporated aluminium computed from the McPeak n, k data [14], 0.904–0.925. Household foil is rougher and oxidised, so this is an upper bound; the grease contact is idealised as uniform.

High-gloss paper.. The paper is wrapped with an air gap and modelled as `polishedbackpainted` with the backing index set to air. The plate keeps its full TIR cone and only the sub-critical fraction crosses the gap to the paper, which returns it specularly with a fitted $R(\lambda) \approx 0.900$–0.955. The reference measurement reports glossy paper and aluminium as the two brightest wrappers, which motivates modelling glossy paper as a slightly lossy mirror rather than a diffuser. This is a model choice: real photo paper has a diffuse component and its reflectance was not measured independently.

Table 1: Principal parameters of the scintillator, WLS fibre and SiPM adopted in the simulation, compared with the values quoted in Ref. [6]. A dash marks a quantity not reported or not modelled. The density and Birks constant are taken from the Geant4 NIST database and Ref. [11], respectively, not from the UPS-92S data sheet.

Parameter	Ref. [6]	This simulation
Scintillator		
Material	UPS-92S (AdvTECH UK)	PVT (`G4_PLASTIC_SC_VINYLTOLUENE`)
Dimensions	$250 \times 250 \times 10$ mm	$250 \times 250 \times 10$ mm
Refractive index	1.58	1.58
Density	1.05 g cm^{-3}	1.032 g cm^{-3}
Light output	50–60% of anthracene	1.0×10^4 MeV^{-1}
Emission peak	425 nm	425 nm
Bulk attenuation length	2000 mm	3.8 m (peak)
Decay / rise time	2.4 ns / —	2.1 ns / 0.9 ns
Birks constant	—	0.126 mm MeV^{-1}
Surface finish	polished	polished
WLS fibre (Kuraray Y-11(200))		
Diameter	1.0 mm	1.0 mm
Cladding	multi-clad	PMMA ($n = 1.49$) + fluoropolymer ($n = 1.42$)
Core index	—	1.59
Absorption / emission peak	430 nm / 476 nm	430 nm / 476 nm
WLS absorption length at 430 nm	—	1.0 mm
WLS decay time	—	7 ns (exponential)
Read-out	4 fibres, one row, 250 mm	4 fibres, 2.5 mm pitch → 1.0 mm bundle pitch
Channel diameter	—	1.1 mm (0.05 mm air annulus)
SiPM (CEFLA ASD-NUV4S-P-40)		
Active area	4×4 mm^2	4×4 mm^2
Cells	9340	—
Peak sensitivity	420 nm	410 nm
PDE at peak	43%	0.431 (digitized)
Epoxy index	1.5115	1.55
Coupling	BC-630 grease	0.1 mm grease ($n = 1.46$)

Table 2: Geant4 optical-surface parameters for the four reflectors, as attached in the code; type and finish names are hyphenated for space, the identifiers use underscores. $R(\lambda)$ is tabulated on a shared ten-point 360 nm to 650 nm grid and quoted as its range; a dash marks a property that is neither set nor used. The bare plate carries no surface object and uses the built-in `glisur/polished` default; the black holder carries an absorbing skin.

	Bare	**TiO_2**	**Al foil**	**Glossy**
Contact	1.2 mm air gap	direct	grease film	air gap
Surface object	none	skin	skin	skin
Type	dielectric–dielectric	dielectric–dielectric	dielectric–dielectric	dielectric–dielectric
Model	glisur	unified	unified	unified
Finish	polished	ground-front-painted	polished-back-painted	polished-back-painted
σ_α	—	0.10 rad	0	0
$R(\lambda)$	—	0.955–0.983	0.904–0.925	0.900–0.955
Backing index	—	—	1.46	1.0003
f_{ss} / f_{sl} / f_{bs}	—	0 / 0 / 0	1 / 0 / 0	1 / 0 / 0
TIR	$\theta > 39.3°$	no	$\theta > 67.5°$	$\theta > 39.3°$

2.3. WLS-fibre clamp (config 4)

Four 1.0 mm-diameter Kuraray Y-11(200) multi-clad fibres [15, 16] run the full 250 mm length of one narrow face in a single row. In the straight section the centre-to-centre pitch is 2.5 mm, so a solid web of clamp material separates the milled 1.1 mm channels. Over the last 41 mm (39 mm ramp plus a 2 mm stub) the fibres converge to a touching bundle with 1.0 mm pitch that presents all four ends to one 4 mm×4 mm SiPM; the far ends are closed by aluminium mirror caps. Each fibre is a Y-11 core ($n = 1.59$) inside a PMMA inner cladding ($n = 1.49$) inside a fluoropolymer outer sheath ($n = 1.42$). The production solid is an analytic, mitered set of `G4CutTubs` legs, one per straight segment of the bent centreline, placed so adjoining legs meet at exact miter planes; this avoids the degenerate facets of the earlier tessellated variant. A 0.10 mm grease film couples the plate face to the clamp, and each fibre protrudes 0.05 mm into that film. The 0.05 mm air annulus between fibre and channel is deliberate: the fibre must not touch the $n = 1.55$ clamp, and the low-index sheath needs the clearance to guide light. The clamp is white 3D-printed plastic, modelled as a Lambertian reflector with 85 % reflectivity on every face. A skin on the mirror caps ensures that light

travelling in the core, not only in the cladding, is turned around at the far end. WLS re-emission uses the digitized Y-11 emission spectrum with an exponential delay of 7 ns; the `G4OpWLS` process is enabled automatically when the geometry is built.

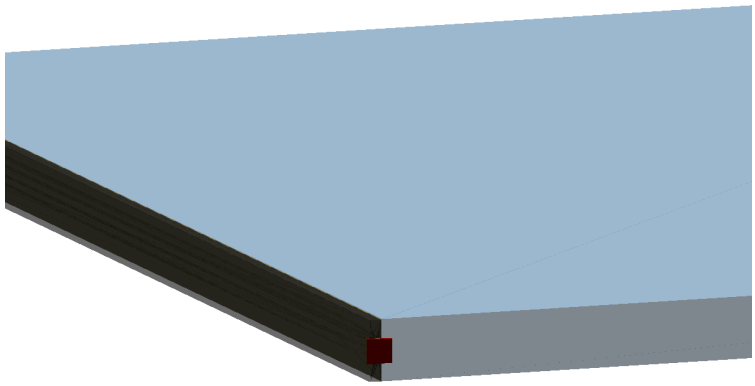


Figure 2: Oblique render of the config-4 assembly: the 250 mm plate with the WLS-fibre clamp on one narrow face and the single read-out SiPM at the end of the converged bundle. The four fibres run the full plate length. The render is geometry-only (no beam) and uses the tessellated visualisation preset; the production geometry uses the analytic mitered solid described in the text.

2.4. Optical inputs and boundary physics

Optical photons are transported by the standard Geant4 optical module, with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes active. Mie scattering is disabled; `G4OpWLS` is enabled only for config 4. Figure 3 shows the spectral tables actually fed to Geant4: the scintillator emission and bulk attenuation, the Y-11 absorption and re-emission,

and the SiPM PDE. The Y-11 absorption peaks at 430 nm and is effectively transparent red-ward of 500 nm; the emission peaks at 476 nm. The emission table is zero red-ward of the absorption onset ($\lambda \geq 488$ nm), but it extends to shorter wavelengths, so a small high-energy tail overlaps the absorption band and mild self-absorption is present by construction.

The boundary process selects a surface from a `G4LogicalBorderSurface` first and a `G4LogicalSkinSurface` second; if neither exists it falls back to a lossless TIR default. A valid `RINDEX` is required on medium 1 always and on medium 2 for a `polished/ground dielectric_dielectric` boundary; `*backpainted` finishes read the backing index from the surface. A missing index produces a silent `NoRINDEX` kill, so every material and backpainted surface carries an explicit table. This is also why the far-end mirror needs a skin: the metal cap has no `RINDEX`, and the uncovered core-to-cap crossing would otherwise be lost.

For the unified model, the reflection weights are `SPECULARSPIKECONSTANT` (f_{ss}), `SPECULARLOBECONSTANT` (f_{sl}) and `BACKSCATTERCONSTANT` (f_{bs}), together with `REFLECTIVITY`. The finish selects the code path: `groundfrontpainted` is a pure Lambertian draw with no TIR; `polishedbackpainted` is a Fresnel/TIR dielectric boundary followed by a backing reflection; and the bare plate uses the default polished dielectric–dielectric boundary. Photodetection is delegated to the `dielectric_metal` grease/SiPM boundary. In `DielectricMetal()`, an absorbed photon is classified as `Detection` only when a uniform draw is below the tabulated PDE; when the boundary-invoke-SD flag is set, the sensitive detector is called only for that outcome. The sensitive detector therefore books one hit per genuine photoelectron and kills the photon; there is no separate random draw in user code.

3. Simulation campaign and observables

3.1. Beam patches and efficiency definition

The primary source is a vertical 4 GeV μ^- beam with a 50 mm square profile. The beam centre is placed at three positions in the report frame,

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

which correspond to slab-local positions $(0, 0)$, $(100, 0)$ and $(100, 100)$; the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$. Each patch has the same area and lies fully on the plate, so the

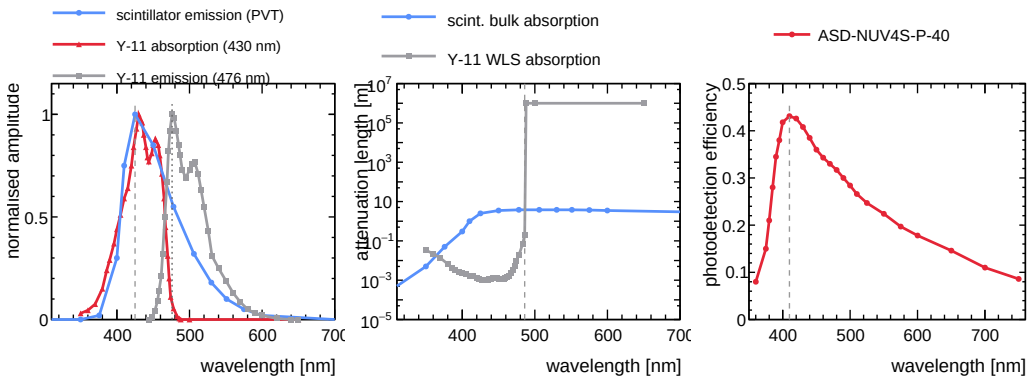


Figure 3: Optical property tables used by the simulation. Left: normalized scintillator emission, Y-11 absorption and Y-11 re-emission. Middle: attenuation lengths on a logarithmic scale, including the scintillator bulk curve and the fibre WLS absorption with 1.0 mm at the 430 nm peak. Right: digitized SiPM PDE, peaking at 0.431 near 410 nm. Markers are the tabulated points and segments are the linear interpolation used by Geant4.

produced-photon denominators are equal. The edge and corner patches are on the bottom edge, opposite the instrumented top face of configs 3 and 4, matching the reference layout. Each (config, wrapper, patch) combination is run for 2500 events, giving $4 \times 4 \times 3 = 48$ files and 120 000 events. Whole-plate runs use a 250 mm$\times$250 mm GPS plane and 25 000 events per (config, wrapper), giving 400 000 events. The total is 520 000 events, or about 1.0×10^{10} tracked optical photons.

The collection efficiency for a set $\mathcal{S}$ of SiPMs is

$$\varepsilon = \frac{\sum_e \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_e N_\gamma^{(e)}}, \tag{1}$$

where $N_{\mathrm{pe}}^{(e,s)}$ is the number of photoelectrons in SiPM s in event e and $N_\gamma^{(e)}$ is the number of optical photons produced in the slab (scintillation plus Čerenkov). The numerator and denominator are summed over events, so ε is a ratio of totals, not a mean of per-event ratios. For the matrix, the sum runs over all SiPMs of the layout and over the events of one patch. For the whole-plate run, it runs over all events and yields the plate-averaged efficiency. Statistical uncertainties are computed from the event-to-event variance of the numerator at fixed denominator (delta method) and cross-checked with

Poisson errors on the photoelectron count; they are quoted in the tables. Ref. [6] quotes a thresholded muon detection efficiency rather than a raw photon-collection efficiency, so the comparison below is restricted to orderings and relative scaling; the absolute ε values are not directly comparable.

The sensitive detector writes one row per (event, SiPM), including zero-detection rows, so the sample is unbiased in the muon entry position. Each row carries the entry position, the event's produced-photon count, the detected count, the first arrival time, the mean and RMS detected wavelength, and variable-length lists of the individual photoelectron arrival times, wavelengths and creating processes (1 = `OpWLS`, 2 = scintillation, 3 = Čerenkov). This allows the efficiency, spectra, timing and process mix to be reconstructed offline at any binning.

3.2. Whole-plate maps and uniformity

The whole-plate events are binned by the primary muon's first-entry position in 5×5 mm cells, giving the field

$$\varepsilon(x, y) = \frac{\sum_{e:(x_e,y_e)\in\text{cell}} \sum_{s\in\mathcal{S}} N_{\text{pe}}^{(e,s)}}{\sum_{e:(x_e,y_e)\in\text{cell}} N_{\gamma}^{(e)}}.$$

A purely radial contrast is not used: the layouts have no continuous rotational symmetry and the plate is square, so a radial average would mix different azimuths and, beyond the inscribed circle, sample only the corner wedges. The uniformity is instead quantified on the plane by the normalized map $\varepsilon/\langle\varepsilon\rangle$ and by the two-dimensional contrast

$$\kappa_{2\text{D}} = \frac{\max_{\text{cell}} \varepsilon}{\min_{\text{cell}} \varepsilon}, \tag{2}$$

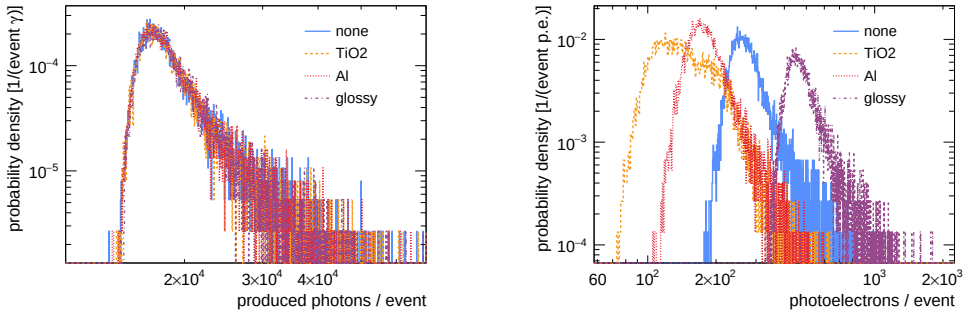
evaluated on a 5×5 grid of 50 mm cells. The coarser grid is used because the 5 mm cells are too sparse for a stable max/min; $\kappa_{2\text{D}}$ is therefore a robust dynamic range, smaller than the raw peak-to-trough range of the fine maps.

4. Results

4.1. Photon yield and photoelectron spectra

Figure 4 shows the per-event produced-photon yield and the detected photoelectron count for config 1 at the centre patch. The mean yield is essentially independent of the wrapper, about 1.9×10^4 photons per event,

as it must be: the wrapper changes transport, not production. The small spread reflects the muon path-length distribution. The photoelectron spectra, in contrast, depend strongly on the wrapper. At the config-1 centre the mean runs from 124 pe/event for TiO_2 and 198 for aluminium to 305 for the air-gap bare plate and 517 for glossy paper. The bare curve is not a near-zero peak but a broad distribution comparable to the reflective wrappers, the direct signature of near-lossless TIR transport. With a PDE of about 0.4 and the geometric acceptance of a side-mounted sensor, a collection efficiency of a few percent is expected; the best wrapper reaches $517/1.9 \times 10^4 \approx 2.7\,\%$.



(a) Produced photons per event. **(b)** Detected photoelectrons per event.

Figure 4: Per-event distributions for config 1 at the centre patch, plotted as probability densities with logarithmic axes. The production yield (left) is wrapper-independent; the detected count (right) depends strongly on the wrapper. The air-gap bare plate is close to the reflective wrappers, not to zero.

4.2. Wrapper ordering and layout scaling

Figure 5 shows the plate-averaged efficiency for all (config, wrapper) pairs, and Table 3 lists the numbers. Glossy paper is the brightest wrapper in every layout. In the three direct-coupled layouts the ordering is

$$\mathrm{TiO_2} < \mathrm{Al} < \mathrm{bare} < \mathrm{glossy},$$

with the bare plate second, not last. In the WLS layout the order changes to

$$\mathrm{Al} < \mathrm{TiO_2} < \mathrm{bare} < \mathrm{glossy},$$

so the aluminium foil becomes the weakest wrapper. The spread from weakest to brightest within a layout is a factor of 2.3–2.9. The efficiency scales roughly

with the photosensitive area: config 1 is highest, config 2 is about 1.9 times lower, config 3 another factor of about 1.9, and config 4 a further factor of 2.1–2.5. The WLS layout collects the fewest photoelectrons because only the fraction of light reaching the fibre face is shifted and guided to the single sensor.

The physical interpretation is straightforward. For the bare plate, the critical angle is 39.3°, so most of the solid angle is trapped by TIR; only the escape cone is lost to the holder, apart from the instrumented pads. For glossy paper, the air gap preserves the same TIR cone and the paper returns part of the escape cone specularly, so glossy is brighter than bare. For aluminium with grease, the critical angle widens to 67.5°: a much larger fraction enters the grease and is reflected by the foil with $R \approx 0.9$, and the repeated Fresnel and reflection losses make it dimmer than the bare TIR guide. For TiO_2, there is no TIR at all; every photon is either diffusely reflected with $R \approx 0.96$ or absorbed, and the random walk loses photons at every bounce. The WLS layout changes the balance because the fibres accept only a small solid angle at the clamp face, and the aluminium foil's high-angle transmission is no longer an advantage.

Table 3: Plate-averaged collection efficiency (all SiPMs, uniform whole-plate irradiation), in units of 10^{-3}. The number in parentheses is the statistical uncertainty in the last digits.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	15.542(12)	9.5318(22)	10.4932(12)	26.4795(12)
Config 2 (4 SiPMs)	8.2579(9)	4.8596(16)	5.4057(9)	13.9865(9)
Config 3 (2 SiPMs)	4.2438(6)	2.4552(15)	2.7254(7)	7.1564(7)
Config 4 (1 SiPM, WLS)	1.7265(2)	1.5891(8)	1.5145(2)	3.4698(3)

4.3. Position dependence

Figure 6 groups the three beam patches by layout, and Table 4 lists the config-1 values with uncertainties. The bare plate and the specular wrappers are nearly flat: for glossy, aluminium and bare the efficiency varies by at most 6 % across the three patches in every layout. Only TiO_2 shows a clear position dependence, with a factor of about 1.75 in config 1. In the four-sided layouts the TiO_2 response is corner > edge > centre; in the single-sided layouts it is centre > edge ≈ corner. The sign follows the distance from the illuminated

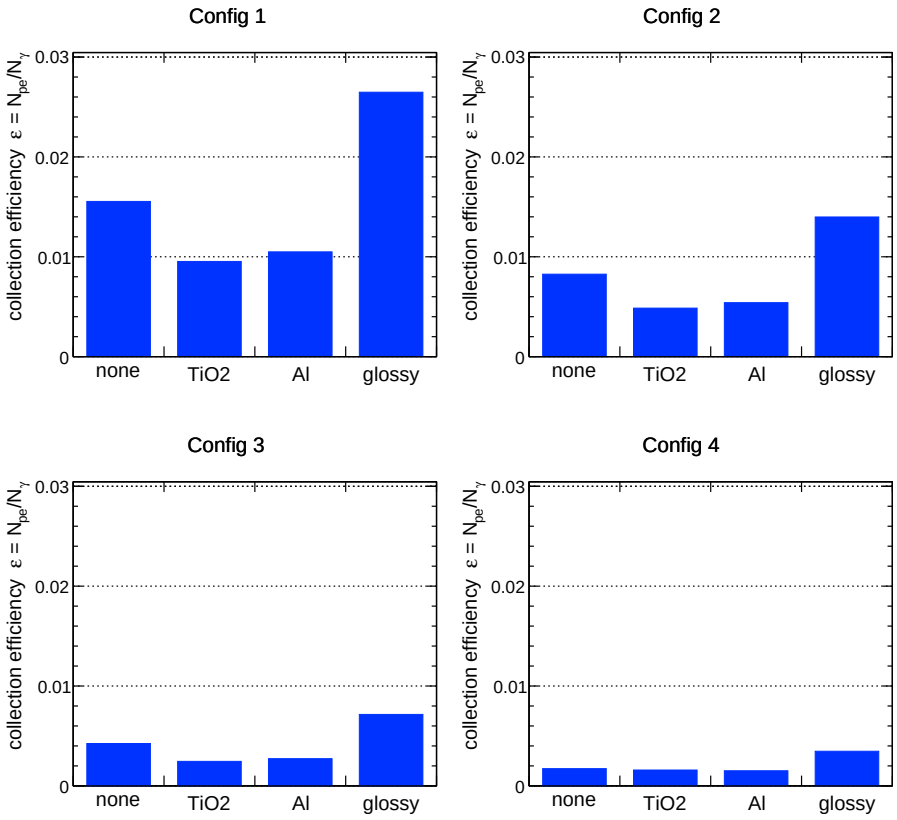


Figure 5: Plate-averaged collection efficiency for the four wrappers and four layouts, from the uniform whole-plate irradiation (25 000 events per panel). Glossy paper is brightest everywhere; the air-gap bare plate is second in the direct-coupled layouts and third in the WLS layout, where aluminium is weakest.

patch to the instrumented face, not a simple radius: in the single-sided layouts the centre patch is closer to the sensors than the edge and corner patches, which lie on the opposite side.

The near-flatness of the bare and specular responses is the signature of a well-mixed cavity: after a few bounces the light no longer remembers where the muon crossed, so the sensor coverage, not the irradiation point, sets the collection. The diffuse paint retains a memory of the source position because each Lambertian bounce randomizes the direction and the light is gradually absorbed; the response is highest where the painted walls can return light efficiently to the sensors.

Table 4: Position dependence of the collection efficiency for config 1 (all SiPMs). Each patch is 50 mm×50 mm and lies fully on the plate. The number in parentheses is the statistical uncertainty in the last digits.

Wrapper	Centre	Edge	Corner
Bare	$1.5485(18) \times 10^{-2}$	$1.4880(19) \times 10^{-2}$	$1.4695(26) \times 10^{-2}$
TiO_2	$6.3687(12) \times 10^{-3}$	$8.0219(15) \times 10^{-3}$	$1.12188(27) \times 10^{-2}$
Al	$1.02503(15) \times 10^{-2}$	$9.7520(16) \times 10^{-3}$	$9.8319(24) \times 10^{-3}$
Glossy	$2.64660(24) \times 10^{-2}$	$2.58179(25) \times 10^{-2}$	$2.56926(30) \times 10^{-2}$

4.4. Whole-plate maps and uniformity

Figure 7 shows the full-plate efficiency maps for config 1. The bare map is nearly flat: the mean within $r < 60$ mm (1.54×10^{-2}) differs from the outer region (1.56×10^{-2}) by under 2 %. Aluminium and glossy are also flat; TiO_2 is the only wrapper with a clear gradient, and it runs the other way, with the centre (6.7×10^{-3}) dimmer than the edge and corner (1.02×10^{-2}), a factor of about 1.5. The same pattern repeats in the other layouts: the bare and specular responses remain nearly flat, while TiO_2 darkens where the source is far from the instrumented face.

The uniformity ranking is the same in every layout,

$$\mathrm{TiO_2} > \mathrm{Al} > \mathrm{bare} \gtrsim \mathrm{glossy},$$

as quantified by κ_{2D} in Table 5. The diffuse paint is the least uniform wrapper, with κ_{2D} growing from 1.90 in config 1 to 8.46 in config 3; the bare plate stays at 1.13–1.47 and the specular wrappers at 1.07–1.79. For the bare

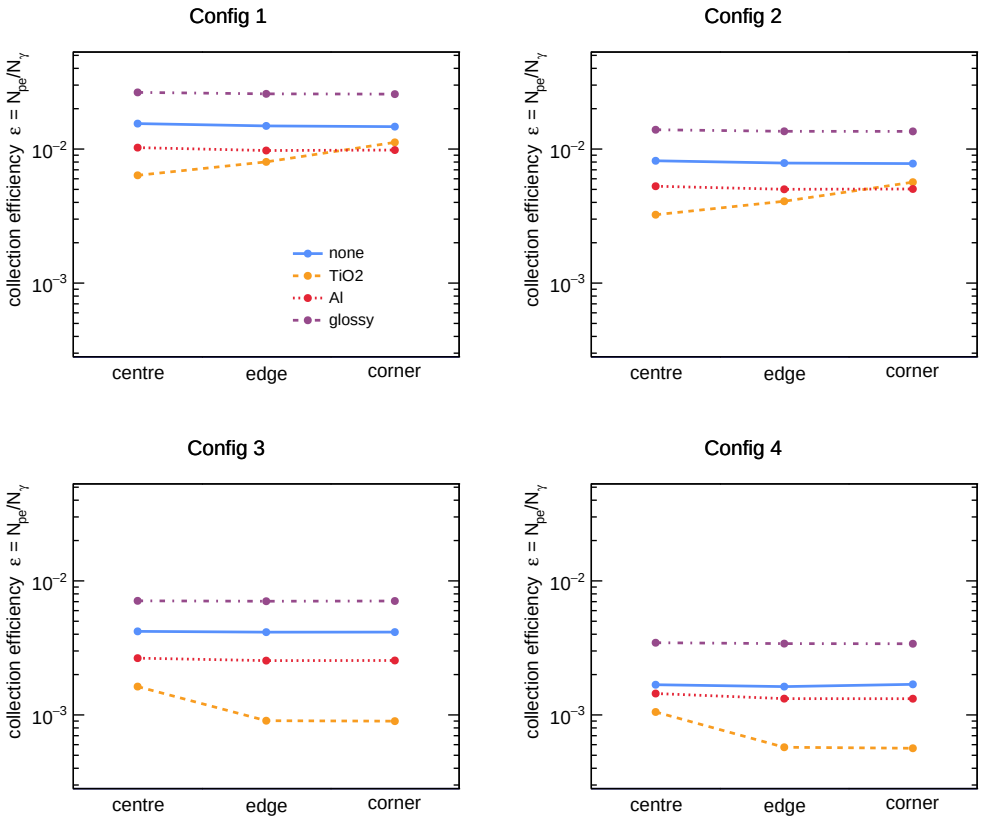


Figure 6: Collection efficiency at the three beam patches, one panel per layout, with the four wrapper curves (logarithmic ordinate). All patches have equal area and lie fully on the plate. The bare and specular wrappers are nearly flat; TiO_2 is the only wrapper with a clear position dependence, and its sign depends on whether the layout is four-sided or single-sided.

and specular wrappers a muon crossing anywhere on the plate is collected within about a factor of two of the most favourable point. For TiO_2 in the single-sided layouts the corner-to-centre penalty approaches an order of magnitude. This is the strongest argument against a diffuse wrapper when position uniformity matters.

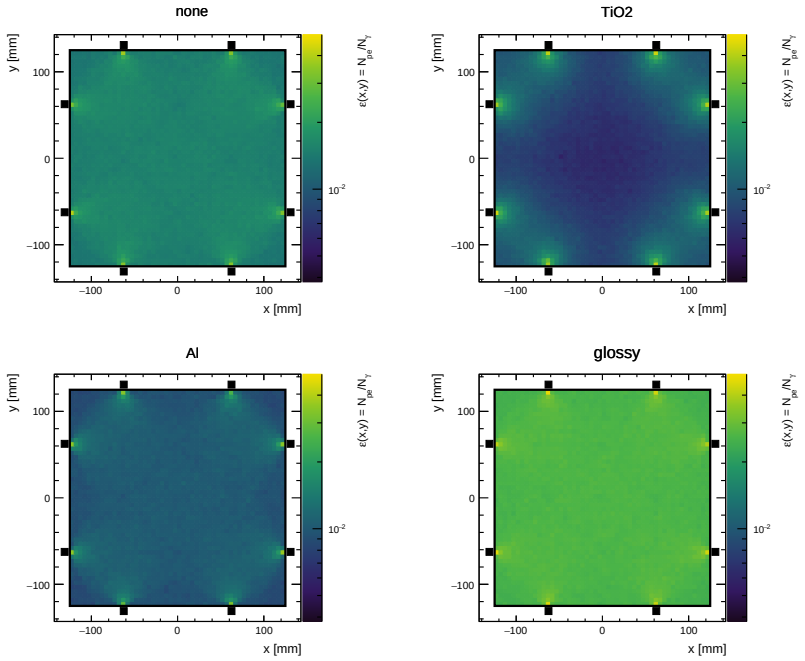


Figure 7: Full-plate collection-efficiency maps for config 1, from 25 000 whole-plate events per panel (5×5 mm cells, logarithmic colour scale shared by the panels). The bare plate and the specular wrappers are almost flat; TiO_2 is the weakest and the only wrapper with a clear gradient, dimmest at the centre and brightest toward the edges and corners. The outline is the 250 mm plate and the squares mark the eight SiPMs.

4.5. Wavelength and timing

Figures 8 and 9 show the detected-light wavelength and arrival-time structure for the glossy whole-plate sample. The direct-coupled layouts (1–3) detect the blue scintillation spectrum with a photoelectron-weighted mean of 456.7 nm to 457.2 nm. Config 4 is green-shifted to 509.6 nm by the Y-11 absorption and re-emission, confirming that the WLS process is active and

Table 5: Full-plate position contrast $\kappa_{2D} = \varepsilon_{max}/\varepsilon_{min}$ over the 50 mm cells of the whole-plate map. The number in parentheses is the propagated Poisson uncertainty in the last digits.

Wrapper	Config 1	Config 2	Config 3	Config 4
Bare	1.129(3)	1.206(4)	1.473(7)	1.361(10)
TiO_2	1.902(7)	2.543(12)	8.460(70)	8.093(81)
Al	1.186(4)	1.329(6)	1.792(11)	1.619(13)
Glossy	1.069(2)	1.119(3)	1.273(5)	1.161(6)

dominates the config-4 spectrum. In every layout the spectrum reddens with arrival time: the mean rises from 451 nm ($t < 5$ ns) to 468 nm ($t > 20$ ns) for configs 1–3, and from 499 nm to 511 nm for config 4. The depleted blue edge of the late light is the signature of self-absorption along the multi-bounce path and, in config 4, of the wavelength-dependent attenuation of the fibre.

The timing is equally informative. Configs 1–3 peak at 3.6 ns to 4.1 ns, with about 30 % of the photoelectrons arriving within 5 ns and 60 % within 10 ns. Config 4 peaks at 9.2 ns, with only 7 % within 5 ns and 33 % within 10 ns. The late tail ($t > 20$ ns) carries 11 % to 13 % of the photoelectrons in configs 1–3 but 25 % in config 4. The integrated mean yields are 520, 275, 141 and 68 pe/event for configs 1–4, consistent with Table 3. The WLS read-out is thus not only less efficient but also slower, which matters for trigger latency and pile-up.

5. Discussion

5.1. The bare-plate discrepancy

The central disagreement with Ref. [6] is the bare plate. Their pulse-height spectra show the bare plate in its black holder as the weakest configuration, whereas our model gives it the second-highest light yield in the direct-coupled layouts. The model is not tuned to force agreement. With a true 1.2 mm air gap, the polished plate is a TIR guide: only the escape cone ($\theta < 39.3°$) is lost, and the trapped light reaches the edge sensors with little bulk attenuation. The measured suppression therefore requires a loss channel that is absent from the nominal construction.

Two candidates are physically plausible. The first is optical contact between the plate and the holder. A real clamped plate touches at scattered

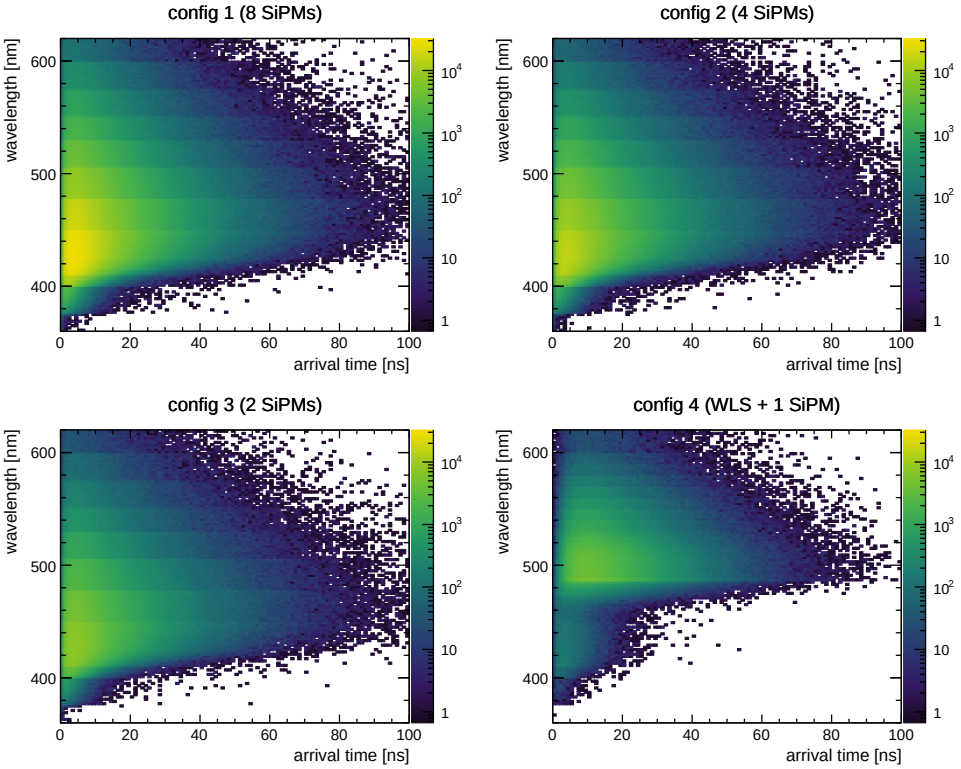


Figure 8: Detected wavelength versus arrival time, one panel per layout (glossy paper, whole-plate irradiation), on a shared logarithmic colour scale. Layouts 1–3 populate a narrow band around 457 nm; config 4 is green-shifted by the Y-11 fibres. The mean wavelength increases with arrival time because the blue edge is preferentially self-absorbed.

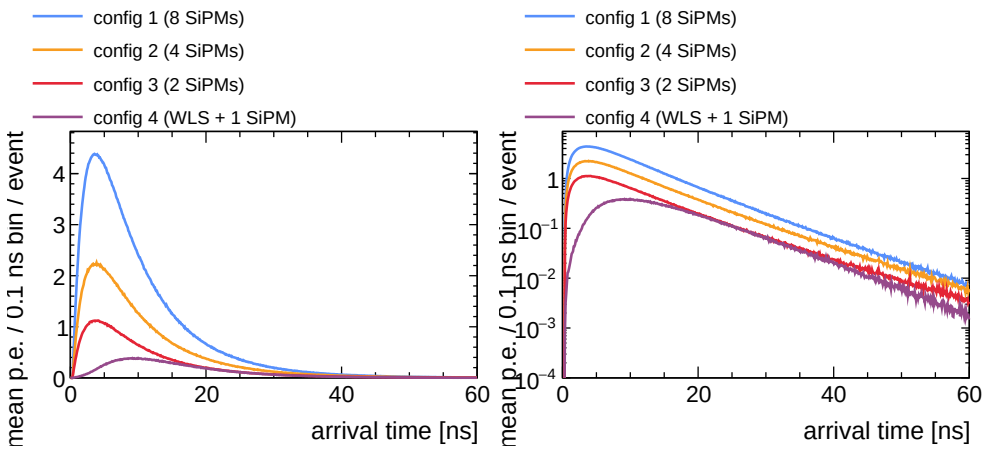


Figure 9: Average arrival-time waveform of the detected photoelectrons, one curve per layout (glossy paper, whole-plate irradiation). The left panel uses a linear ordinate and the right a logarithmic one. Configs 1–3 rise within a fraction of a nanosecond of one another; the WLS layout peaks about 5 ns later and retains a long tail.

high spots; each contact frustrates TIR and removes trapped light. The second is surface roughness. A ground or frosted face scatters light out of the TIR cone at every bounce. Absorption alone is unlikely to explain the suppression: the visible bulk attenuation length of PVT is metres, so the plate would have to be far more absorbing than the nominal material to fall below TiO_2. Erasing TIR instead requires a substantially rougher surface than the nominal polished finish. We did not perform a dedicated roughness scan here. We regard the air-gap model as the defensible reading of the stated construction (a polished plate simply set in a holder) and flag the discrepancy rather than tune the surface to fit the measurement. The practical implication is that the plate–holder interface, not the wrapper alone, is a first-order design variable. A controlled experiment that varies the contact area or the surface finish would test this hypothesis directly.

5.2. Design guidance

Several conclusions are robust across layouts. Glossy paper is the brightest and most uniform wrapper, but it is a model choice as much as a measurement: the paper is treated as a slightly lossy specular mirror, and real photo paper has a diffuse component. Aluminium foil with grease is intermediate in the

direct-coupled layouts and weakest in the WLS layout; it is also the most sensitive to assembly quality, because wrinkles and trapped air reduce its effective reflectance. TiO_2 paint is the weakest and least uniform wrapper in the direct-coupled layouts; its diffuse recycling helps only when the source is near a painted edge, and it produces the largest corner-to-centre penalty in the single-sided layouts. A bare polished plate is an excellent light guide only if it is genuinely decoupled from its holder; if contact or roughness is unavoidable, the bare configuration can be the worst choice, as the reference measurement suggests.

The layout scaling is equally clear. Eight sensors on four faces give the highest yield; four sensors on four faces halve it; concentrating two sensors on one face halves it again; and the single-SiPM WLS clamp is about a factor of two lower still. The WLS layout trades absolute yield for a single read-out channel and a large effective collection area. Its green-shifted, slower light is a disadvantage for timing but may be acceptable for a trigger with a loose time window. A cost-normalized comparison would require the electronics and dark-count model that this work does not include.

5.3. Limitations

The absolute efficiencies are model-dependent. They are fixed by the assumed light yield, the Birks constant, the PDE curve, the grease coupling and the tabulated reflectivities. The TiO_2 and glossy-paper reflectance curves are representative fits, not measurements of the actual materials, and the diffuse lobe of the paint is modelled as ideal Lambertian. The bare plate assumes a perfect air gap and no contact; this is the largest geometric uncertainty. The WLS fibre uses a nominal 1.0 mm absorption length at 430 nm; a systematic band of 0.3 mm to 3 mm is available in the code but not propagated here. The SiPM is modelled as an ideal photocathode with the datasheet PDE; gain, dark counts, crosstalk, afterpulsing, electronics thresholds and temperature are not included [17]. The primary beam is a monoenergetic 4 GeV muon, not the cosmic spectrum, and the matrix patches sample only three positions. The statistical uncertainties are quoted, but a full systematic budget is beyond the scope of this work.

The geometry also contains deliberate simplifications. Configs 3 and 4 are both placed on the same narrow face, whereas Ref. [6] places config 3 on the top face and config 4 on the left face, i.e. on adjacent faces. This simplification keeps the single-sided layouts directly comparable and opposite the edge/corner beam patches, but it means the config-3/config-4 comparison

is not an exact reproduction of the reference geometry. The WLS fibre pitch and convergence length are chosen to fit the 250 mm plate and the single SiPM; the reference gives less detail on the clamp interior. These choices affect the absolute config-4 yield, but not the conclusion that the WLS read-out is slower and less efficient per produced photon.

6. Conclusion

We have built a Geant4 optical model of a polished plastic scintillator read out by side-mounted SiPMs and used it to study four layouts and four wrapper treatments at three beam positions and over the whole plate. The model uses the unified surface model, digitized scintillator and Y-11 tables, a digitized SiPM PDE and boundary-process photodetection, and it reproduces the measured wrapper ordering in part: glossy paper is the brightest and most uniform wrapper, and TiO_2 is the weakest and most position-dependent wrapper in the direct-coupled layouts. The bare plate is the central discrepancy: with a true air gap it is a near-lossless TIR guide and the second-brightest wrapper, not the weakest. The measured suppression therefore points to a loss channel at the plate–holder interface—optical contact or surface roughness—rather than to the wrapper alone. The WLS layout collects the fewest photoelectrons and is about 5 ns slower, but it uses a single sensor and a large effective area. The results provide quantitative guidance for wrapper and layout choice and identify the plate–holder interface as the main ingredient missing from the present model.

CRediT authorship contribution statement

J. Turko: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence this work.

Data availability

The simulation code, macros and analysis scripts are available from the author on reasonable request. The data that support the findings of this study are available from the corresponding author upon reasonable request.

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